

```
#include <Servo.h>

// Motor A (LEFT) connections
int enA = 3;
int in1 = 4;
int in2 = 5;

// Motor B (RIGHT) connections
int enB = 8;
int in3 = 6;
int in4 = 7;

//Payload connection
int servoPin = 9;
int angle = 0;


#define pauseTime 1000
#define turnTime 400
#define straightTime 3000
#define joltTime 20
#define rightMotorspeed 100
#define leftMotorspeed 101
#define rightTurnspeed 0
#define leftTurnspeed 150


Servo myServo;


void setup() {
    // Set all the motor control pins to outputs
```

```
pinMode(enA, OUTPUT);  
pinMode(enB, OUTPUT);  
pinMode(in1, OUTPUT);  
pinMode(in2, OUTPUT);  
pinMode(in3, OUTPUT);  
pinMode(in4, OUTPUT);
```

```
// Turn off motors - Initial state
```

```
digitalWrite(in1, LOW);  
digitalWrite(in2, LOW);  
digitalWrite(in3, LOW);  
digitalWrite(in4, LOW);
```

```
// Attach the servo to the pin
```

```
myServo.attach(servoPin);
```

```
}
```

```
//Function Execution
```

```
void loop() {
```

```
    delay(10000);
```

```
    drive10Feet();
```

```
    delay(pauseTime);
```

```
    leftTurn90Degrees();
```

```
    delay(pauseTime);
```

```
    drive10Feet();
```

```
    delay(pauseTime);
```

```
leftTurn90Degrees();  
delay(pauseTime);  
drive10Feet();  
delay(pauseTime);  
leftTurn90Degrees();  
delay(pauseTime);  
drive10Feet();  
delay(pauseTime);  
//Calling Payload Drop  
dropPayload();  
delay(pauseTime);  
}
```

```
//Mission 1-A 10 Ft. Square
```

```
//Drive 10 Ft. in a straight line at 1 ft/s
```

```
void drive10Feet(){  
  // Turn on motors  
  digitalWrite(in1, LOW);  
  digitalWrite(in2, HIGH);  
  digitalWrite(in3, HIGH);  
  digitalWrite(in4, LOW);  
  
  analogWrite(enA, 255);  
  analogWrite(enB, 255);
```

```
delay(joltTime);
```

```
analogWrite(enA, leftMotorspeed);
```

```
analogWrite(enB, rightMotorspeed);
```

```
//Run for 10 ft
```

```
delay(straightTime);
```

```
// Now turn off motors
```

```
digitalWrite(in1, LOW);
```

```
digitalWrite(in2, LOW);
```

```
digitalWrite(in3, LOW);
```

```
digitalWrite(in4, LOW);
```

```
}
```

```
//Perform a 90 degree turn
```

```
void leftTurn90Degrees() {
```

```
    // Turn on motors
```

```
    digitalWrite(in1, LOW);
```

```
    digitalWrite(in2, HIGH);
```

```
    digitalWrite(in3, LOW);
```

```
    digitalWrite(in4, HIGH);
```

```
    analogWrite(enA, 255);
```

```
    analogWrite(enB, 255);
```

```
delay(joltTime);
```

```
// Speed analog values for motors
```

```
analogWrite(enA, leftTurnspeed);
```

```
analogWrite(enB, rightTurnspeed);
```

```
//delay for turn
```

```
delay(turnTime);
```

```
// Now turn off motors
```

```
digitalWrite(in1, LOW);
```

```
digitalWrite(in2, LOW);
```

```
digitalWrite(in3, LOW);
```

```
digitalWrite(in4, LOW);
```

```
}
```

```
// Payload Drop Function
```

```
void dropPayload() {
```

```
    // Sweep the servo from 0 to 180 degrees
```

```
    for(angle = 0; angle <= 110; angle += 22) {
```

```
        myServo.write(angle); // Set the servo position
```

```
        Serial.print("Angle: ");
```

```
        Serial.println(angle);
```

```
        delay (5);
```

```
}
```

```
delay (pauseTime);
```

```
// Sweep back from 180 to 0 degrees
```

```
for(angle = 110; angle >= 0; angle -= 22) {
```

```
  myServo.write(angle);
```

```
  Serial.print("Angle: ");
```

```
  Serial.println(angle);
```

```
  delay(5);
```

```
}
```

```
delay (pauseTime);
```

```
}
```